

1.

(a)

(b)

2.

The Poisson distribution for a discrete $k, k = 0, 1, 2, \dots$ and real parameter λ is

$$P(k|\lambda) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

(a)

Find the mean of k .

By definition, we have that

$$\begin{aligned} E[k] &= \sum_{k=0}^{\infty} k \cdot P(k|\lambda) \\ &= \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= 0 + \sum_{k=1}^{\infty} \lambda^k e^{-\lambda} \frac{k}{k!} \\ &= e^{-\lambda} \sum_{k=1}^{\infty} \lambda^k \frac{1}{(k-1)!} \end{aligned}$$

Simplifying by substituting $j = k - 1$:

$$\begin{aligned} &= e^{-\lambda} \sum_{j=0}^{\infty} \lambda^{j+1} \frac{1}{j!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ &= \lambda e^{-\lambda} \cdot e^{\lambda} \end{aligned}$$

$$\boxed{E[k] = \lambda}$$

(b)

Find the variance of k .

We need to essentially find $E[k^2]$ here. We can quickly ascertain that directly substituting k^2 will make this hard to simplify and cancel out the k terms in both the numerator and factorial denominator, so we can instead solve for $E[k(k-1)]$:

$$\begin{aligned} E[k(k-1)] &= \sum_{k=0}^{\infty} k(k-1) \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= \sum_{k=0}^{\infty} e^{-\lambda} \frac{(k-1) \cdot \lambda^k}{(k-1)!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{(k-2)!} \\ &= 0 + 0 + e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!} \end{aligned}$$

Simplifying by substituting $j = k - 2$:

$$\begin{aligned} &= e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+2}}{j!} \\ &= \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ E[k(k-1)] &= \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2 \\ E[k^2 - k] &= \lambda^2 \\ E[k^2] - E[k] &= \lambda^2 \\ E[k^2] &= \lambda^2 + E[k] = \lambda^2 + \lambda \end{aligned}$$

We can now solve for variance:

$$\begin{aligned} \text{Var}(k) &= E[k^2] - E[k]^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ \boxed{\text{Var}(k) &= \lambda} \end{aligned}$$

(c)

Find the mode of k .

(d)

Assume two categories C_1 and C_2 , equally probably *a priori*, distributed with Poisson distributions and $\lambda_1 > \lambda_2$. What is the Bayes classification decision?

(e)

What is the Bayes error rate?

3.

(a)

(b)

4.